Programming Merit Badge Workshop

By Levi Johanon

Safety

Digital Safety Video

https://filestore.scouting.org/filestore/YPSAT/YT%20Mod1%20Final%20Master%20Small.mp4

How to Reduce Eye Strain

History

History

https://www.youtube.com/watch?v=K5yv7dY17EA

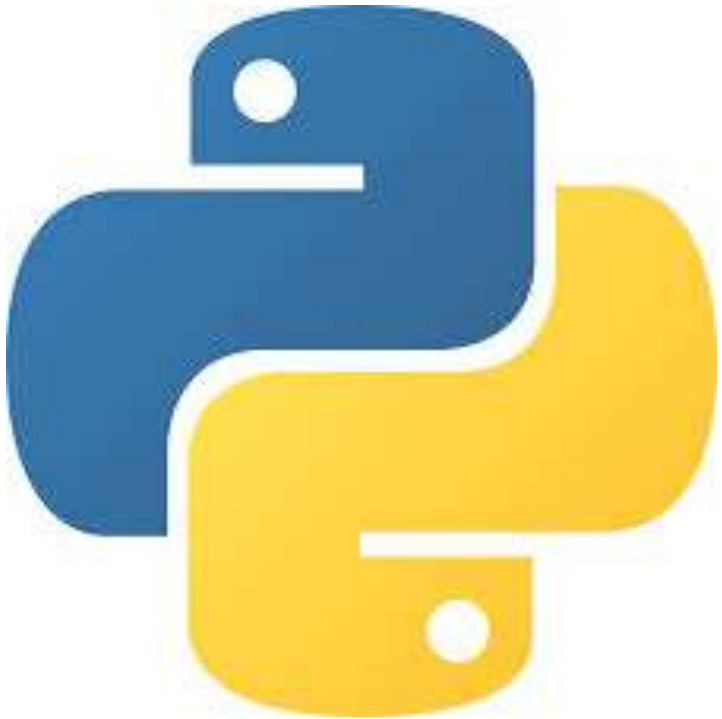
Different Programming Languages

Python

Industries: Data Science, AI/ML, Web Development, Finance, Education

Why: Python’s simple syntax and extensive libraries (e.g., NumPy, TensorFlow, Django) make it ideal for rapid development, prototyping, and data analysis. It’s a go-to for machine learning and scientific computing.

Example Code: print("Hello, World!")



Java

Industries: Enterprise Software, Android Development, Banking, E-Commerce

Why: Known for stability, portability, and scalability. Java is widely used in large systems and Android apps due to its robust ecosystem and JVM (Java Virtual Machine).

Example Code:

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

C

Industries: Systems Programming, Embedded Systems, Operating Systems, Compilers, Telecom, Automotive

Why: C provides direct memory access, deterministic performance, and minimal runtime overhead. It’s the backbone of operating systems, kernels, drivers, firmware, and embedded devices. Its portability and long-standing standardization make it foundational across critical infrastructure and low-level software.

Example Code:

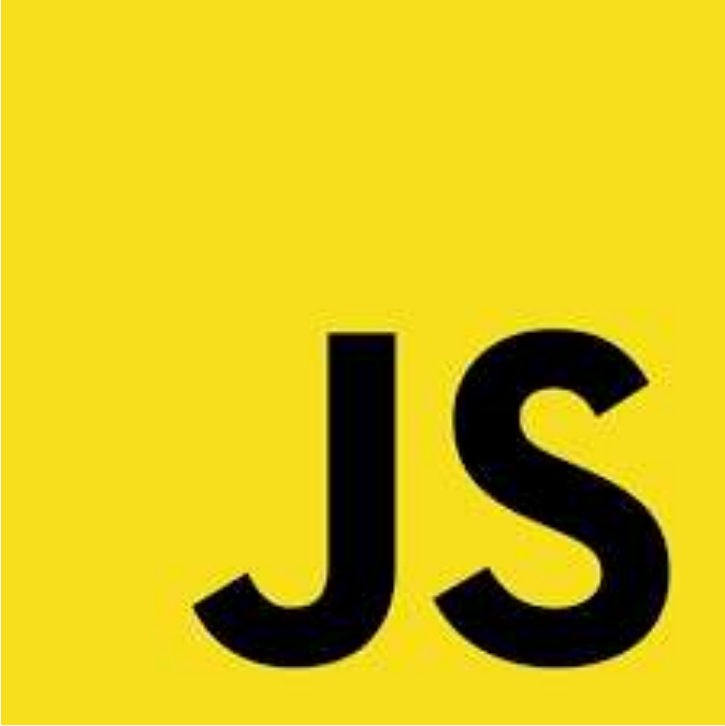
```
#include <stdio.h>

int main() {
    printf("Hello, World!\n");
    return 0;
}
```

Javascript

Industries: Web Development, E-Commerce, Mobile Apps, SaaS

Why: JavaScript is the standard for front-end development of webpages, enabling dynamic websites and web apps. With Node.js, it's also used on the server-side, offering full-stack development capabilities.

Example Code: console.log("Hello, World!");



SQL

Industries: Data Analytics, Finance, Healthcare, E-Commerce, Enterprise Software

Why: SQL is the standard language for querying and managing relational databases. Its declarative style makes it ideal for handling large datasets, performing analytics, and powering data-driven applications across nearly every industry that relies on structured data.

Example Code: SELECT 'Hello, World!' AS message;

Intellectual Property

4 Types

- Copyright – Protects the expression of ideas, not the ideas themselves. Automatically applied and it lasts for the life of the author.
- Patent – Protects the idea, it needs to be applied for and lasts for 20 years.
- Trademark – Protects brand identifiers. Can last forever.
- Trade Secret – Not officially recognized or protected, kept through secrecy and NDAs. Can be reverse engineered

Licensed vs Owned Software

- Licensed Software – You do not own the product; you are simply using it under certain conditions by the owners.
- Examples: Microsoft Office, Photoshop, Steam Games
- Owned Software – You have full ownership of the source code. You can use, distribute, and resell as you please.
- You typically only own software you own or buy the rights to, even open-source code is typically licensed.

Freeware, Open-source, & Commercial

- Freeware is a program that is free to download and use, but you don’t have the code. Example - Zoom
- Open-source is a program where you can see the source code and share it. You can improve it if you want. Example – Firefox or this workshop.
- Commercial software is software that is made to be sold and isn’t owned by you. Example - Photoshop

Coding Language 1: Arduino (C/C++)

Inspiration

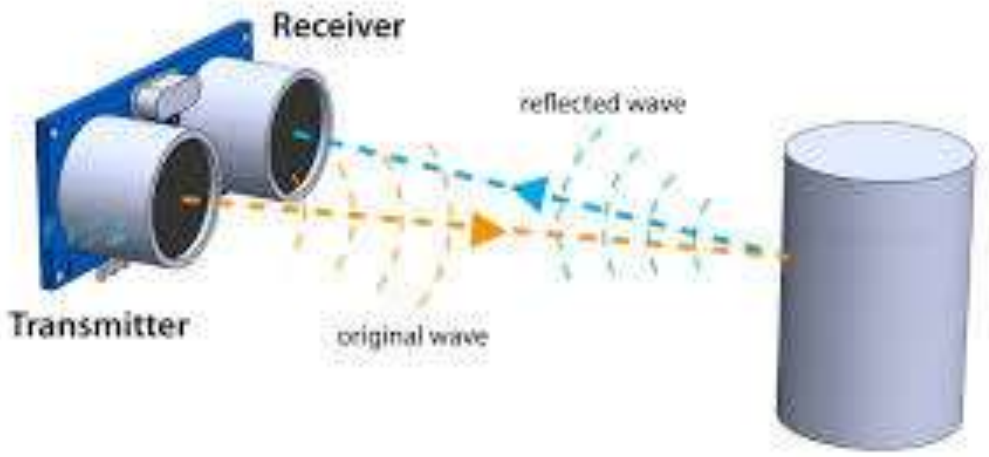
https://www.youtube.com/watch?v=mC1ikwQ5Zgc&t=207s
Skip to 3:02

Resources

- Servo Motor
- Ultrasonic Sensor
- Ring Light

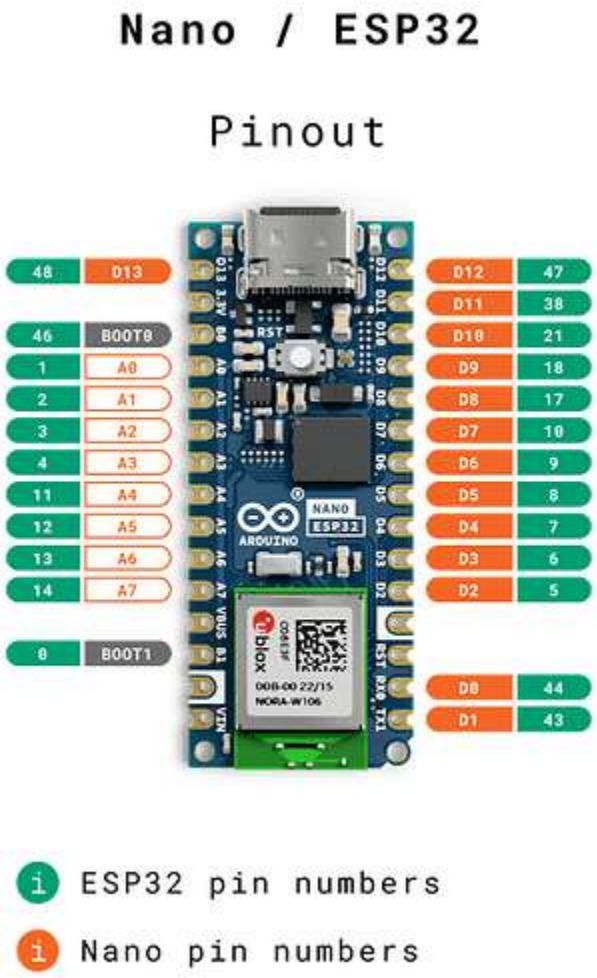
How ultrasonic sensor works

- Send a brief high signal to Trigger pin
- The echo pin will be high until the vibration bounces back.
- Goal: convert echo pin response in microseconds to distance
- Speed of sound: 343m/s = 34300 cm/s = 0.0343 cm/us,
- Formula: d = 0.0343 * time / 2



Arduino

- Setup
- Loop



Basic Syntax

#define NAME PIN

pinMode(pin,OUTPUT/INPUT);

Create Variables:

Variable_type variable_name = initial_value;

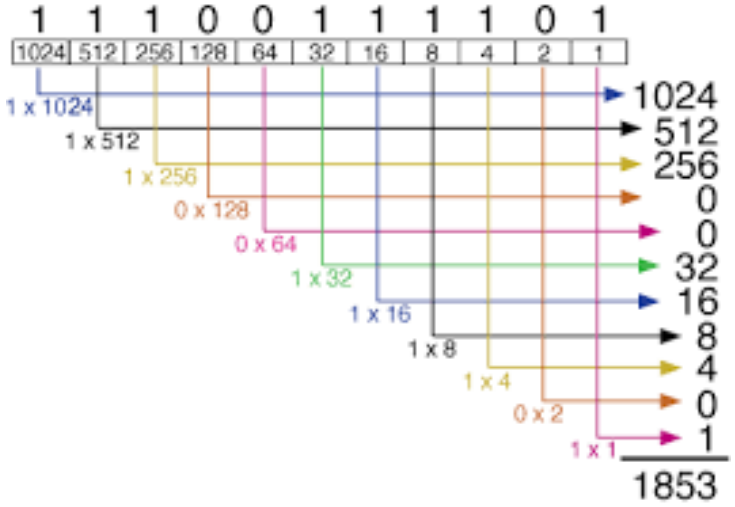
long x = 0; -long is a large number used which will be used for storing microseconds

float y = 5.7; -float is a number with decimal points, it will store the distance

Coding Basics

Binary

- Information on computers is stored in binary
- A bit is one digit of information, a true or false
- N bits can represent 2^N possible pieces of information
- 64 bits can store a number between 0 and 18 quintillion

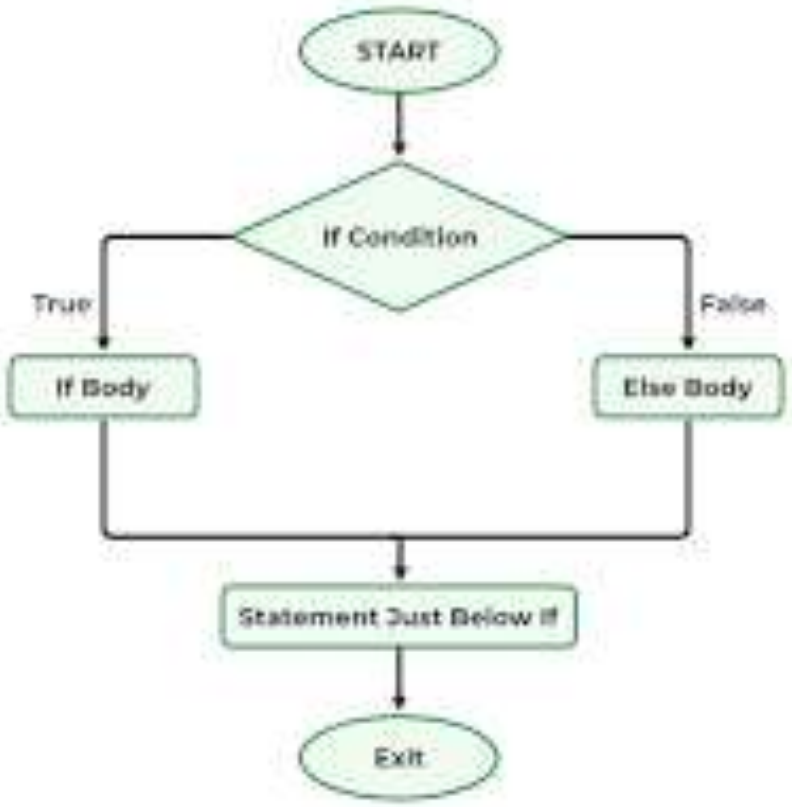


Data Types

- Whole Numbers (Integers)
- Boolean (True or False)
- Characters (ASCII)
- Floating point numbers

0 011 0000	A 100 0001	N 100 1110	a 110 0001	n 110 1110	. 010 1110
1 011 0001	B 100 0010	O 100 1111	b 110 0010	o 110 1111	, 010 1100
2 011 0010	C 100 0011	P 101 0000	c 110 0011	p 111 0000	! 010 0001
3 011 0011	D 100 0100	Q 101 0001	d 110 0100	q 111 0001	? 011 1111
4 011 0100	E 100 0101	R 101 0010	e 110 0101	r 111 0010	' 010 0111
5 011 0101	F 100 0110	S 101 0011	f 110 0110	s 111 0011	(010 1000
6 011 0110	G 100 0111	T 101 0100	g 110 0111	t 111 0100	) 010 1001
7 011 0111	H 100 1000	U 101 0101	h 110 1000	u 111 0101	- 010 1101
8 011 1000	I 100 1001	V 101 0110	i 110 1001	v 111 0110	" 010 0010
9 011 1001	J 100 1010	W 101 0111	j 110 1010	w 111 0111	space 010 0000
	K 100 1011	X 101 1000	k 110 1011	x 111 1000	
	L 100 1100	Y 101 1001	l 110 1100	y 111 1001	
	M 100 1101	Z 101 1010	m 110 1101	z 111 1010	

If Statements

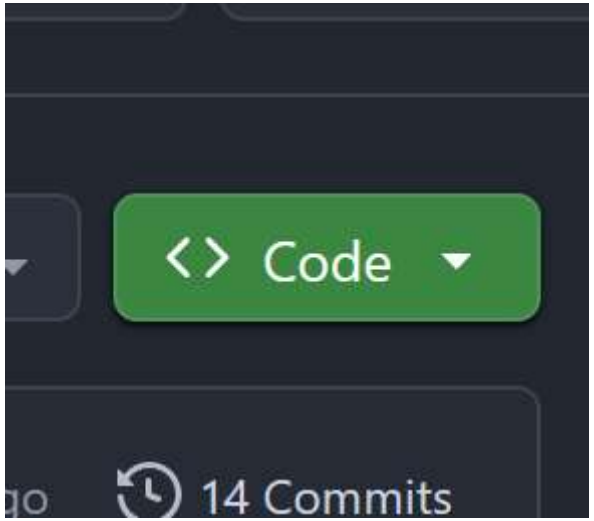



```c
int n = 10;

if (n > 5)
{
    printf("%d is greater than 5", n);
}
else
{
    printf("%d is less than 5", n);
}

return 0;
```

Back to Arduino

Download Files from Github

https://github.com/ljohanon06/Scouting_America



Arduino Cloud IDE

https://cloud.arduino.cc/

TODO 1: Add Pin Definitions

Replace TODO with a name for the trigger and echo pins, they should be all caps. You will use this name every time in the code later on.

```
//TODO: Add pin definitions
#define /*TODO*/ A5
#define /*TODO*/ A6

#define LED_PIN D2
```

TODO 2: Create variable names

Decide what to name your variables. There is two common syntaxes:

camelCase and under_scores

Choose one and stick to it

```
//TODO: Name variables
long /*TODO*/; //Variable will store the length of the echo pulse in microsecond
float /*TODO*/; //Variable will store the calculated distance to the object
```

TODO 3: Complete Setup Code

Set your trigger and echo pins as either inputs or outpus

```
void setup() {
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  pinMode(/*TODO*/,/*TODO*/); //Set the trigger pin as an input or output
  /*TODO*/                    //Set the echo pin as an input or output

  Serial.begin(9600);
}
```

TODO 4: Finish the function

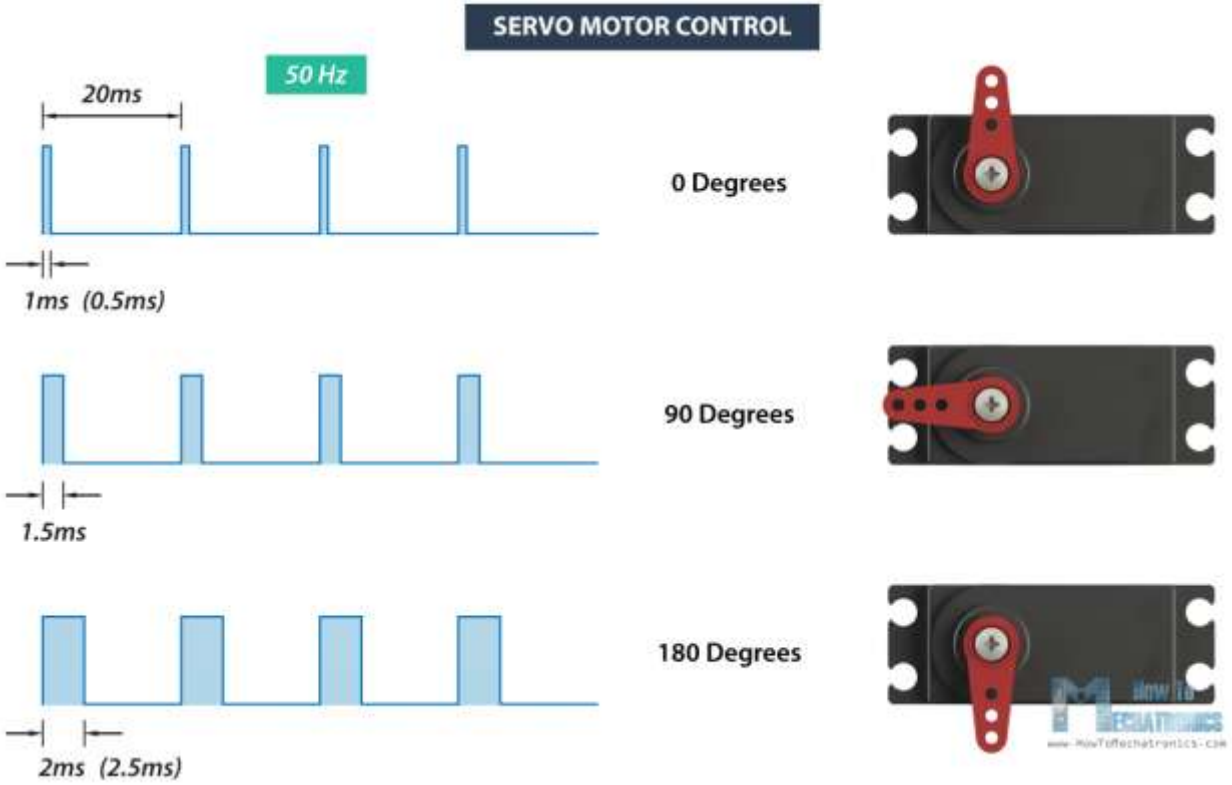
```
void measure_ultrasonic(){
    //TODO: Send a pulse on the trigger pin
    digitalWrite(/*TODO*/, /*TODO*/);
    delayMicroseconds(2);

    digitalWrite(/*TODO*/, /*TODO*/);
    delayMicroseconds(10);
    digitalWrite(/*TODO*/, /*TODO*/);

    //TODO: Measure response on echo pin
    /*TODO*/ = pulseIn(/*TODO*/, HIGH);

    //TODO: Calculate the distance to the object from the duration that the echo pin is high
    
}
```

TODO 5: Finish the loop

```
void loop() {
  if(count % 100 == 0){
    measure_ultrasonic();
    //TODO: Print the distance to the object
    Serial.print("Distance: ");
    Serial.print(/*TODO*/);
    Serial.println(" cm");

    //TODO: Check if the object is in a certain area
    if(/*TODO*/ < /*TODO*/ || /*TODO*/ > /*TODO*/){
      digitalWrite(LED_PIN,HIGH); //Object is gone
    }else{
      digitalWrite(LED_PIN,LOW);
    }
  }
```

Servo Code

How Servos Work



TODO 1: Names!

Create names for everything

```
#include <ESP32Servo.h>

#define /*TODO*/ 8 //Name for the servo pin

#define /*TODO*/ D4 //Name for the button pin

volatile bool /*TODO*/ = false; //Variable name storing if the button has been pressed

unsigned long count = 0;

Servo /*TODO*/; //Name for your servo
```

TODO 2: Setup

```
void IRAM_ATTR /*TODO*/(){
    /*TODO*/ = true;
}

void setup() {
  /*TODO*/.setPeriodHertz(50);              //Servos run at 50 Hz
  /*TODO*/.attach(SERVO_PIN, 500, 2400);

  /*TODO*/.write(180);

  pinMode(/*TODO*/, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(/*TODO*/),/*TODO*/,/*TODO*/);

}
```

TODO 3: Loop

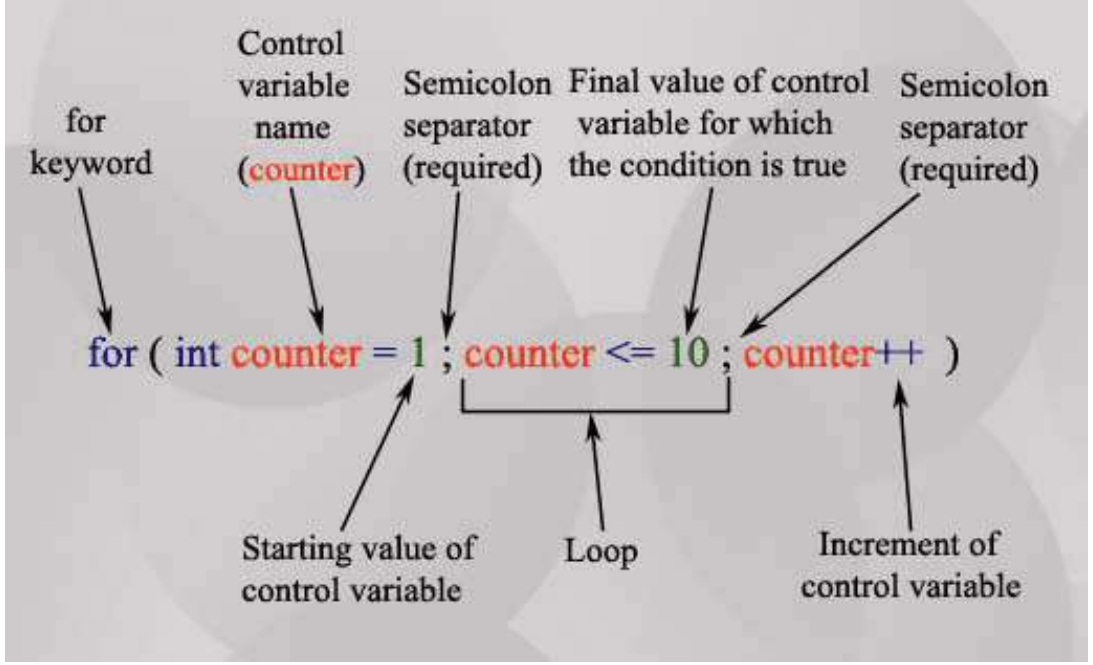
```
void loop() {
  if(/*TODO*/){
    //TODO: move servo to 0 degrees
  }

  count++;
  delay(10);
}
```

Ring Code

TODO: Variables

```
#include <Adafruit_NeoPixel.h>

#define /*TODO*/ D4 //Button pin

#define /*TOTO*/ 17 //Name the NeoPixel Led Ring pin
#define LED_COUNT 12

Adafruit_NeoPixel /*TODO*/(LED_COUNT, /*TODO*/, NEO_GRB + NEO_KHZ800); //Ring is transmitted Gree

unsigned long count = 0;

//TODO: Add the button pressed variable and function from earlier

void alarm_ring();
void calm_ring();

void setup() {
  /*TODO*/.begin();    //Initializes ring
  /*TODO*/.show();     //Pushes current LED setup to ring, as nothing has been setup yet, LED is b

  //TODO:Setup the Button like earlier

}
```

For loops


for keyword
Control variable name (counter)
Semicolon separator (required)
Final value of control variable for which the condition is true
Semicolon separator (required)
for (int counter = 1 ; counter <= 10 ; counter++)
Starting value of control variable
Loop
Increment of control variable

Count Divider

Integer Division: The Decimal Part is cut off

5/2 = 2, 13/10 = 1, 12345/200 = 61

Integer Modulus: Provides the remainder

5%2 = 1, 13%10 = 3, 12345%200 = 145

```
int count_div = (count/a)%b;
```

a: How often your led changes

b: How many states you want

Functions

```
int red = a;
int green = b;
int blue = c;

ring.fill(ring.color(red,green,blue),start,end);
ring.fill(red,green,blue,start,end);
ring.setPixelColor(ring.color(red,green,blue),pixel_num);
```

Example 1

```
void alarm_ring(){
  int count_div = (count/100)%2;

  if(count_div == 0){
    ring.fill(ring.Color(100,0,0),0,12);
  }else{
    ring.fill(ring.Color(0,100,0),0,12);
  }
}
```

Example 2

```
void alarm_ring(){
  int count_div = (count/100)%10;

  for(int i = 0; i<12; i++){
    int red = 10 * ((count_div+i) % 12);
    ring.setPixelColor(red,0,0,i);
  }
}
```

Example 3

```
void alarm_ring(){
  long count_div = (count/50)%10;

  int g_b = 90-count_div*10;
  ring.fill(ring.Color(100,g_b,g_b),0,12);
}

void calm_ring(){
  long count_div = (count/10)%100;

  int r_b = 90-count_div;

  ring.fill(ring.Color(r_b,100,r_b),0,12);
}
```

Example 4

```
void alarm_ring(){
  long count_div = (count/50)%3;

  switch(count_div){
    case 0: ring.fill(ring.Color(100,0,0),0,4); break;
    case 1: ring.fill(ring.Color(0,100,0),4,8); break;
    case 2: ring.fill(ring.Color(0,0,100),8,12); break;
  }
}
```

TODO: Fill in alarm and calm rings

```
void alarm_ring(){
  /*TODO*/
}

void calm_ring(){
  /*TODO*/
}
```

TODO: Loop

```
void loop() {
  if(/*TODO*/){
    /*TOTO*/();
  }else{
    /*TODO*/();
  }
  /*TODO*/.show();

  count++;
  delay(10);
}
```

Putting it all together

TODO: Fill in the blanks

```
#include <Adafruit_NeoPixel.h>
#include <ESP32Servo.h>

#define /*TODO*/ 17
#define LED_COUNT 12    //Number of leds on ring

#define /*TODO*/ 8

#define /*TODO*/ A5
#define /*TODO*/ A6

#define /*TODO*/ D2
#define /*TODO*/ D4

Adafruit_NeoPixel /*TODO*/(LED_COUNT, /*TODO*/, NEO_GRB + NEO_KHZ800); //Ring is tran
Servo /*TODO*/;

unsigned long count = 0;

long /*TODO*/;
float /*TODO*/;

void alarm_ring();
void calm_ring();

void measure_ultrasonic();

volatile bool /*TODO*/ = false;
bool /*TODO*/ = false; //Variable showing if the object has been taken

void IRAM_ATTR /*TODO*/(){
    /*TODO*/ = true;
}
```

TODO: Fill in more blanks

```
void setup(){
    //Ring Initialize

    //Servo Initialize

    //Led/Button initialize

    //Ultrasonic Initialize
}
```

TODO: Fill more blanks

```
void alarm_ring(){
  /*TODO*/
}

void calm_ring(){
  /*TODO*/
}

void measure_ultrasonic(){
    /*TODO*/
}
```

TODO: Trigger alarm mode

```
if(count % 10 == 0 && !/*TODO*/){
    measure_ultrasonic();

    if(/*TODO*/){
        /*TODO* = /*TODO*/;
    }
}
```

TODO: What to do in Alarm Mode

```
if(/*TODO*/ || /*TODO*/){
    //TODO: What happens in alarm mode
}else{
    //TODO: What happens in calm mode
}
/*TODO*/.show();
```

Full Testing!!

Coding Language 2: Python

Your Input

10000 Employees

Mixed Part and Full Time Workers

Part Time salary is Hours-Wage

Last Column is Age

```
Constancy Mills | Part | 21-37 | Female | 23
Guenevere Hartman | Full | 47300 | Female |
Theresa Sosa | Full | 70700 | Female | 32
Donnie Snow | Part | 32-11 | Female | 29
Dreddy Chambers | Part | 17-19 | Male | 36
Gloriana Roman | Full | 63167 | Female | 30
Marcella Watson | Full | 46578 | Male | 31
Cherilynn Cortez | Part | 26-21 | Male | 44
Dinny Cortez | Full | 62808 | Male | 29
Allyce Fulton | Part | 30-13 | Male | 64
Glad Brady | Part | 33-13 | Male | 26
Colene Fisher | Full | 40669 | Male | 48
Shayna Dotson | Full | 70312 | Female | 25
Krystle Stevenson | Full | 73243 | Male | 27
Evangelina Bowers | Part | 27-14 | Male | 44
Yolanthe Miranda | Part | 20-102 | Male | 33
Giselle Bowman | Part | 30-18 | Female | 21
Ambur Sellers | Part | 17-12 | Male | 41
Chrystal Cooper | Part | 31-12 | Female | 35
Ethyl Mcgee | Full | 66636 | Male | 31
Winona Nunez | Full | 46935 | Male | 46
Sadella Holland | Full | 59353 | Male | 30
Nelly Marshall | Part | 22-41 | Male | 34
```

Your Goal: What is the average age

Coding Language 3: Java

Your Goal: Find the average salary